Publicaciones:

An approximate model of a multilayer wire-on-tube condenser operating with R134a and R600a: experimental validation and parametric analysis, Case Studies in Thermal Engineering, 2021.

Experimental study of temperature erosion tests on bidirectional coated and uncoated composites materials. Materials research express, 2020.

Parametric analysis of the thermal behavior of a single-channel solar chimney, Solar Energy, 2020.

Insights into the transition from plug to slug flow in a horizontal pipe: An experimental study. Chemical Engineering Research and Design, 2020.

Numerical Study of the Double Diffusion Natural Convection inside a Closed Cavity with Heat and Pollutant Sources Placed near the Bottom Wall, energies, 2020.

Numerical Study of the Distribution of Temperatures and Relative Humidity in a Ventilated Room Located in Warm Weather, Computer Modeling in Engineering & Sciences, 2020.

Experimental study of an earth to air heat exchanger (EAHE) for warm humid climatic conditions, Geothermics, 2020.

Wind speed variability study based on the Hurst coefficient and fractal dimensional analysis. Energy Science and Engineering, 2019.

Characteristics of churn and annular flows in a large diameter vertical riser. International Journal of Multiphase Flow, 2019.

Development of a solar calorimeter for the thermal evaluation of glazing samples, BUILDING PHISICS, 2019.

Diameter scaling correlations for near atmospheric air-water slug velocities in horizontal pipe. Journal of Petroleum Science and Engineering, 2019.

Experimental investigation of interfacial waves in stratified liquid-liquid flows in horizontal pipelines: Characteristics and pressure gradients. Journal of Petroleum Science and Engineering, 2019.

Overview of low GWP mixtures for the replacement of HFC refrigerants: R134a, R404A and R410A, International Journal of Refrigeration, 2019.

Techno-economic feasibility study of small wind turbines in the Valley of Mexico metropolitan area. Energies, 2019.

Wind turbulence intensity at la Ventosa, Mexico: A comparative study with the IEC61400 standards. Energies, 2018.

Annular liquid film thickness prediction in a vertical 180° return bend. Experimental Thermal and Fluid Science, 2018.

Interrogating flow development and phase distribution in vertical and horizontal pipes using advanced instrumentation. Chemical Engineering Science, 2018.

Numerical Study on the Effect of Distribution Plates in the Manifolds on the Flow Distribution and Thermal Performance of a Flat Plate Solar Collector, energies, 2018.

Comparative Account of Phenolics, Antioxidant Capacity, α-Tocopherol and Anti-nutritional Factors of Amaranth (Amaranthus hypochondriacus) Grown in the Greenhouse and Open Field, INTERNATIONAL JOURNAL OF AGRICULTURE & BIOLOGY, 2018.

Harmonic modelling of the wind turbine induction generator for dynamic analysis of power quality. Energies, 2018.

Transient behavior of CO2 in the internal heat exchanger during the start-up of the transcritical refrigeration system, Science and Technology for the Built Environment, 2017.

Applications of artificial neural network (ANN) method for performance prediction of the effect of a vertical 90 bend on an air–silicone oil flow. Journal of the Taiwan Institute of Chemical Engineers, 2017.

The control and maintenance of desired flow patterns in bends of different orientations. Flow Measurement and Instrumentation, 2017.

Comparative analysis of a concentric straight and a U-bend gas cooler configurations in CO2 refrigeration system, International Journal of Heat and Mass Transfer, 2017.

Wind speed forecasting using the NARX model, case: La Mata, Oaxaca, México. Neural Computing and Applications, 2016.

Numerical study of the effect of buoyancy on conjugate heat transfer in simultaneous turbulent flow in parallel pipelines, International Journal of Heat and Mass Transfer, 2016.

Double Diffusion Numerical Simulation for an Aired Room with Inner Pollutant Sources, Springer International Publishing Switzerland, 2016.

Capítulos de Libro:

Abrasive Wear Performance of Fe2B Layers Applied on Steel Substrates. Friction, Lubrication and wear, 2019.

Estudio de iluminación en un aula de educación superior para eficiencia energética, Perspectivas de la ingeniería para el Desarrollo Sustentable, 2018.

Participación en Congresos:

Modal Analysis on a Wind Turbine Blade Based on a Wind Tunnel Experiments. *Topics in Modal Analysis & Testing*, The Society for Experimental Mechanics, Inc. 2021.

Aplicación del Conocimiento:

Patente: Máquina de movimiento reciprocante para pruebas de desgaste. No: en trámite, 2020.